

Prahari

Federation middleware for the state's CCTV, not another VMS.

**We ship 200-byte sighting rows, not video.
That is the whole architecture.**

Model Hybrid — M1 foundation, M2 surface, M3 core **Mandatory analytic** ANPR **Grid** ~50 heterogeneous cameras → 80,000

Stack open-source, vendor-neutral **Licence** Apache-2.0

DEMO VIDEO youtu.be/8lQsuXeH150

SOURCE github.com/Priyanshu-byte-coder/prahari

Solution presentation · deliverable 1 of 3. The High-Level Design and the workflow / integration diagram are submitted alongside it, and every number in this deck names the command that produced it.

THE HYBRID

It is not any one of the four reference models

Model 1 is the foundation the brief says it must be. Model 3 is the architecture built on top of it. From Models 2 and 4 we take a technique and an ambition — and reject Model 4's central video store outright.

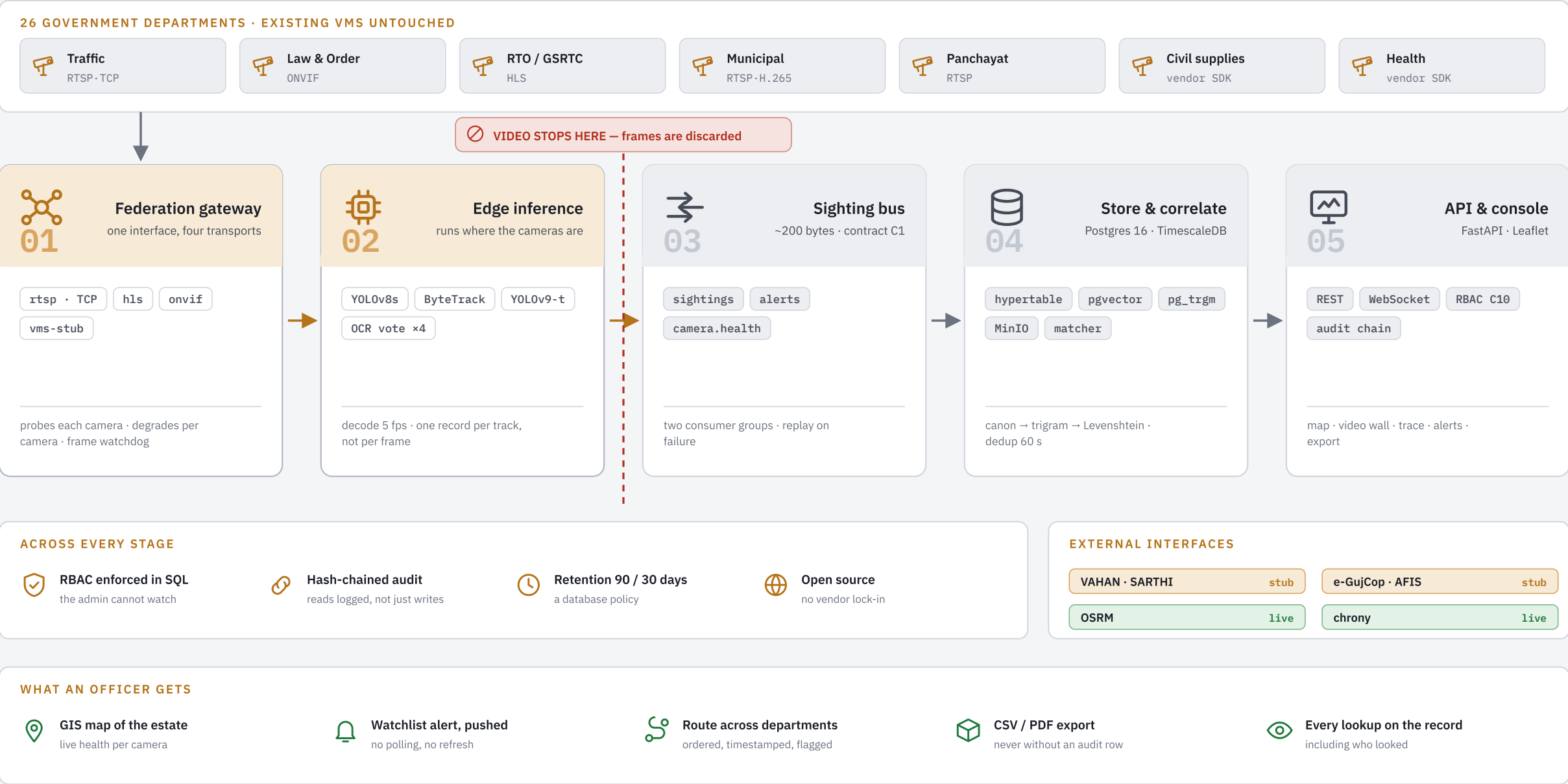
MODEL	WHAT WE TAKE	WHAT WE REJECT	WHERE IT LIVES
M1 Registry + GIS	The camera registry as the spine of record; every asset on a map with a real coordinate	Its assumption that one department owns every camera	cameras table, camera_geo.json
M2 Direct-attach viewing	The direct RTSP / WHEP / HLS view — no transcode hop, no per-vendor console	M2 as an <i>architecture</i> : it forbids the federation layer we depend on	MediaMTX + the wall
M3 Federation middleware	The structure — drivers behind one interface, one event bus, one API	Its heavyweight per-site install	gateway drivers, Redis Streams
M4 Central VMS	Cross-department correlation and watchlist analytics	Central recording of 80,000 streams — the arithmetic is slide 09	matcher, alerts, access_grants

One line for the jury: federation middleware with pluggable drivers — three drivers live, the departmental-VMS driver interface-complete pending vendor credentials.

Prahari — system architecture

Hybrid M1 + M2 + M3 · open source · no vendor lock-in

 **Video never leaves the department**
A ~200-byte sighting is the only thing that does.



WHAT AN OFFICER GETS

 **GIS map of the estate**
live health per camera

 **Watchlist alert, pushed**
no polling, no refresh

 **Route across departments**
ordered, timestamped, flagged

 **CSV / PDF export**
never without an audit row

 **Every lookup on the record**
including who looked

DRIVERS

Four transports live, one interface-complete — and we say which

DRIVER	PROTOCOL	STATUS	CAMERAS CARRIED ON THE SENTINEL GRID
hls	HLS over HTTPS	LIVE	30 of 30. Every grid camera is pulled this way today.
rtsp	RTSP / TCP	LIVE	0 — the grid answers 401 on 8554 for our source IP. Proven on our own feed.
mediamtx	RTSP / WHEP / HLS	LIVE	Carries our own camera in the recorded demo.
onvif	ONVIF Profile S + events	LIVE	0 — the sandbox exposes no ONVIF endpoint. Discovery and PTZ tested against an ONVIF simulator.
vms-stub	vendor VMS SDK	INTERFACE-COMPLETE	0 — awaiting vendor credentials.

The honest thing, said out loud: the fourth driver is an interface, not a claim. On our own network port 8554 is filtered, so the gateway probes each camera and degrades to HLS by itself. Heterogeneity is exactly what the brief tests — that fallback is the demo, not an excuse.

THE POINT The grid permits exactly one of our four transports, and the platform found that out by itself — per camera, at startup, and recorded in `camera:transport:<id>`. Nothing downstream noticed.

THE MANDATORY ANALYTIC

The pipeline, and the record it produces

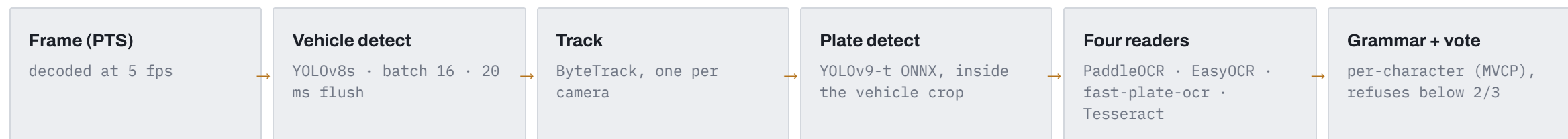


Plate detection runs on the *vehicle crop*, not the frame: a 20 px plate in a 1920 px frame is ~7 px after letterboxing; inside a 300 px crop it is a readable band. Grammar ($\wedge[A-Z]\{2\}[0-9]\{2\}[A-Z]\{1,3\}[0-9]\{4\}\$,$ BH series too) is applied before the vote, so the vote spends itself only on genuine ambiguity.

```
# transcribed from `python services/worker/selftest.py` – injected GJ 25 BJ 8377
{
  "sighting_id": "01M1W3M62K...", "camera_id": "SELFTEST-000",
  "plate_text": "GJ25BJ8377", "plate_conf": 0.989, "plate_band": "CONFIRMED",
  "pts_first": 1725100841.20, "ts_source": "frame_pts", // never now()
  "vehicle_class": "car", "crop_uri": "s3://..."
}
```

Read back exactly, **0.68 s end to end** against a 3.0 s budget.

MODEL CARD

Four bands, never averaged into one number

Banded output — 40 tracks, 120 crops

BAND	N	SHARE	WRONG
CONFIRMED	38	95.0%	0
PROBABLE	0	0.0%	0
POSSIBLE	2	5.0%	0
NONE	0	0.0%	0

Exact match **95.0%** of all tracks and **100%** of the tracks it agreed to name. CER 0.050.

Refusal rate 5.0% — on the slide, because leaving it off is how 70% becomes 95%.

Source: `scripts/accuracy_report.py`, synthetic set, 2026-09-04. The hand-labelled Gujarat set is open work, and this deck does not pretend otherwise.

Why a vote, and not the best single reader

READER ALONE	EXACT	CER
paddleocr	79.2%	0.121
easyocr	70.0%	0.221
fast-plate-ocr	55.0%	0.241
tesseract	42.5%	0.257
fused vote	95.0%	0.050

Four architectures disagree in different places. The vote is worth 16 points over the best of them — and it is what turns a wrong read into a refusal rather than a confident lie.

THE HARD CASE, MEASURED

A wide-area camera is not an ANPR camera. We measured how far short it falls.

Cameras on the Sentinel grid put a number plate into 20–30 px of frame. That is the regime the brief means by “challenging and varying CCTV conditions”, so it is the regime we benchmarked — rather than reporting a number from a clean plate and hoping.

The case the grid actually presents: a 20 px plate

PLATE HEIGHT, NIGHT	LR → OCR	SINGLE-IMAGE SR	OURS — MFSR + MVCP
16 px	0.0%	0.0%	5.0%
20 px	0.0%	0.0%	65.0%
24 px	0.0%	15.0%	95.0%

There is a cliff just under 20 px, and we admit it

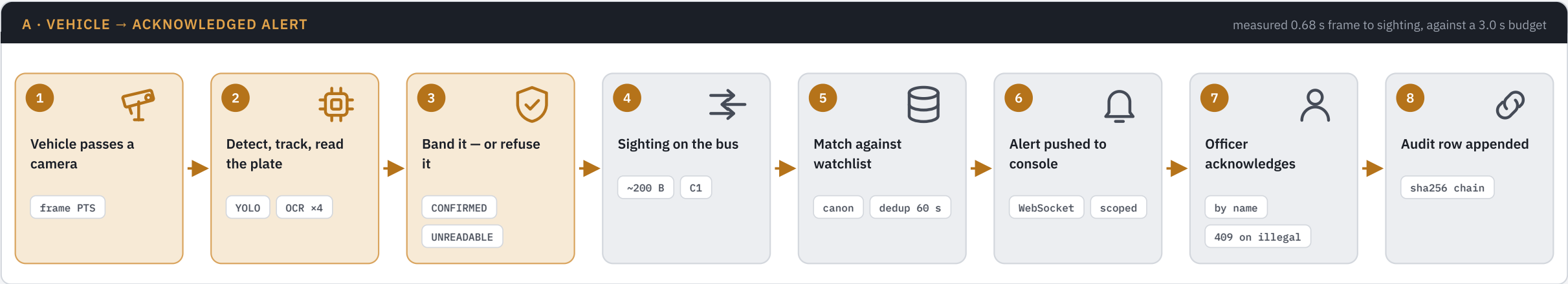
The published reference for this regime — UFPR-SR-Plates, median plate height 18–21 px — reports **1.7–2.2%** from a low-resolution crop straight to OCR, and **42–45%** with super-resolution plus a per-character vote. That is the bar, and it is why a wide-area camera cannot be treated as an ANPR camera.

A plate face is about 65% glyph, so 20 px of plate is ~13 px of glyph — the bottom of what CRNN and SVTR recognisers were trained on. Below ~10 px the pipeline refuses, and the wall says so on the tile.

WHY THIS MATTERS TO A PROCUREMENT It separates a siting problem from a model problem. A plate under ~20 px will not read on any software; the answer is a camera at the right place, and the platform saying so on the tile instead of printing noise.

Prahari — workflow & integration

Two paths. Both end on the same hash-chained audit log.



REAL-TIME ALERTS

Banded matching, an audited state machine

Matching is banded, not binary: exact `plate_norm`, then `plate_canon` (confusion classes collapsed), then weighted edit distance ≤ 1.0 . Watchlist categories: stolen vehicles, wanted / missing persons, blacklisted vehicles, suspect lists.



Every transition is written to the audit log with a user and a reason. Repeated sightings of one vehicle bump a count on the open alert — twenty alerts for one event is how an operator learns to ignore them.

MEASURED Pipeline half — frame in to sighting published: **0.68 s** against a 3.0 s budget (`services/worker/selftest.py`). Dedup holds one alert per (entry, camera) for 60 s; an illegal transition is HTTP 409 and writes nothing.

ROUTE RECONSTRUCTION

A registration number in. A defensible route out.

A judge types a vehicle registration; the map draws that vehicle's hops across the grid with timestamps and locations, snapped to roads via OSRM.

What it returns

- Ordered hops: camera, coordinate, timestamp
- Implied speed on each leg
- CSV export, audited

What it refuses

- A hop needing 180 km/h through a city is flagged **IMPLAUSIBLE**, not quietly drawn
- A route that cannot lie to an officer is worth more than one that always answers

Expected output for the jury: complete route, timestamped and location-wise, plus the live watchlist cross-referencing and auto-alerts running alongside it.

RBAC + AUDIT

Enforced in SQL. Hiding a button is not access control.

The department-scope matrix is enforced in the database with row-level security as the backstop — not in the UI. A district viewer's query simply returns fewer rows; a system administrator may not see live data at all.

Every action is logged

- Every export and every live view writes an audit row
- Cross-department access needs a case number, a 72 h cap and target-department approval
- The log is hash-chained; a tamper breaks the chain

Integration-ready

- APIs shaped for VAHAN, SARTHI, eGujCop (CCTNS), AFIS, NAFIS
- Documented, open protocols — no lock-in on cameras, VMS, analytics or storage

VERIFIED GET /api/admin/audit/verify → **728 rows, ok: true, first_broken_seq: null**. And the row the matrix exists for: **SYSTEM_ADMIN is refused live data, alerts and route** — the most privileged account in the system cannot watch anybody.

THE ARITHMETIC, PRINTED SO THE JURY CAN CHECK IT

Four orders of magnitude, in four multiplications

Centralised video — the M4 approach we reject:

```
80,000 cameras × 2 Mbps (H.264, 720p, 5 fps) = 160,000 Mbps ≈ 160 Gbps
160 Gbps × 86,400 s/day                      ≈ 1.7 PB/day
                                              ≈ 26 PB at ~15-day retention
```

Metadata — what we actually ship:

```
~200 bytes per sighting row ([C1])
80,000 cameras × ~1,000 sightings/camera/day ≈ 80 M rows/day
80 M × 200 B                                ≈ 16 GB/day raw
                                              ≈ 2-3 GB/day compressed (Timescale)
```

26 PB → 16 GB/day. That is the entire case for edge inference.

Scale plan also covers: central / regional / edge compute, GPU sizing, low-bandwidth operation, hot / warm / cold storage by retention, HA, backup, DR, cybersecurity, and a capex + opex estimate.

EVERY CLAIM ON THIS DECK, AND THE COMMAND THAT CHECKS IT

Nothing here is a screenshot of an intention

COMMAND	RESULT
pytest tests/ -q	372 passed, 6 skipped
python scripts/verify_stack.py	35 / 35 checks passed end to end against the running stack
python services/worker/selftest.py	SELFTEST OK — plate read back exactly, 0.68 s against a 3.0 s budget
PRAHARI_INTEGRATION=1 pytest tests/test_integration.py	6 / 6 — clip → worker → real Redis stream → persister → matcher → alert → route
GET /api/admin/audit/verify	728 rows, ok: true
python scripts/accuracy_report.py	95.0% exact, 0 confidently-wrong reads
python scripts/preflight.py	13 / 13 — and every failure prints the command that repairs it

100,366 sightings have gone through the live pipeline on this machine. The brief says *not a simulation, not a proof of concept* — so the repository ships the checks, and a judge can re-run every line above. Open gaps are named on the roadmap slide rather than hidden here.

WHAT IS BUILT, WHAT IS NEXT

Honest about the line between the two

NOW

Four live transports · ANPR with a four-reader banded vote · watchlist alerts with an audited state machine · route reconstruction with plausibility flags · RBAC + hash-chained audit · GIS console + 30-camera video wall with live detector overlays.

NEXT 30 DAYS

The VMS driver against real vendor credentials · detectors fine-tuned on labelled Gujarat footage (+ TensorRT) · re-ID as PROBABLE corroboration · crowd, wrong-way and loitering analytics.

THEN

Face matching under a separate legal authorisation · ANPR-grade camera-siting guidance — a plate under 100 px wide is a siting fault, not a model fault.

Operational impact: one console for 26 departments, a stolen vehicle traced across the state in seconds instead of phone calls, and every lookup on the record — so the system is trusted enough to be used.

The fusion refuses to guess below a two-thirds majority. We would rather report nothing than report the wrong vehicle to a police officer.

DEMO VIDEO youtu.be/8lQsuXeH150

SOURCE, APACHE-2.0 github.com/Priyanshu-byte-coder/prahari